

Bodily Condition Estimation of Archery Learners Using Interoceptive Accuracy

Zexiong Gao
t19295zg@sfc.keio.ac.jp
Keio University
Japan

Satoki Hamanaka
hamachu@sfc.keio.ac.jp
Keio University
Japan

Yuki Inui
t20072yi@sfc.keio.ac.jp
Keio University
Japan

Kentaro Goto
t19335kg@sfc.keio.ac.jp
Keio University
Japan

Hiroo Bekku
bekkuzer@sfc.keio.ac.jp
Keio University
Japan

Wataru Sasaki
wataru@ht.sfc.keio.ac.jp
Keio University
Japan

Tadashi Okoshi
slash@sfc.keio.ac.jp
Keio University
Japan

Jin Nakazawa
jin@sfc.keio.ac.jp
Keio University
Japan

ABSTRACT

Previous studies have shown that not only player skill but also psychological factors contribute to improving archery scores. Therefore, in addition to improving learners' posture, an archery learning support system is required to improve psychological coping skills, such as mental and physical condition management skills. This study propose a method for estimating the condition of archery learners in order to provide appropriate instruction in accordance with their condition. In previous studies of condition estimation, input data are usually objective while label data are usually subjective. Such bias between data is considered as a partial reason of estimation accuracy reduction. Judgments of conditions are subjective and individually different. Subjects' emotion recognition abilities(ERA) are the individual differences in the judgment of emotions, a part of condition in our study. We utilized interoceptive accuracy to objectively evaluate subjects' ERA. Based on archery movement data collected from 10 subjects, we compared the baseline method and the proposed method with the addition of objectively evaluated ERA. The results showed that the latter method, which took into account individual differences of ERA among subjects, estimated condition better than baseline method. We believe that proposed condition estimation method, which pays regard to users' mental and physical condition, is not only useful in sports scenarios, but also necessary for supporting human well-being based on better understanding of user context.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Ubiquitous and mobile computing**.

KEYWORDS

machine learning, multi-modal, emotion estimation, interoceptive accuracy

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1 INTRODUCTION

There are a variety of factors that influence sports performance. Besides factors including postures and skills that is easy to recognize, physical and mental conditions are also crucial but often forgotten. In the coaching context, identifying the problematic factors, including postures, skills and such conditions, is important to improve their performance.

Today's archery coaching also requires instructions that are appropriate to the condition of the athletes. Optimal emotional condition for archery performance has been identified in the previous study [20]. We believe that instructions based on the estimation of condition including emotion, stress and tiredness while practicing archery, are necessary. Furthermore, such condition estimation method is a basic and crucial part for supporting human well-being based on better understanding of user context.

Subjects' self-reports of condition are often used as labels for condition estimation [8]. Such labels are from a subjective viewpoint, while video and biometric information used as input data are objective. The bias between the subjective label data and the objective input data differs from person to person. Such individual

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differences in the judgments of emotion are described as emotion recognition ability(ERA) [13]. We believe that the problem of reduced estimation accuracy due to such bias can be improved by evaluating the subject's ERA objectively.

Therefore, this study proposes a method for estimating the condition of archery learners that takes into account their ERA (Figure 1). Specifically, to reduce the bias between objective data and subjective evaluation of emotions, we designed a proposed model inputting variables of ERA additionally. We utilized interoceptive accuracy(IAC) to evaluate subjects' individual ERA. As the baseline model, a model with no variables of ERA, but only video and biometric information, was designed in our study. We compared the baseline method and the proposed method based on the data of 10 subjects, and verified whether IAC can contribute to improve the accuracy of the condition estimation.

Reducing the bias between objective data and subjective data in condition estimation by objectively evaluating the subjects' ERA, the contributions of this study are as follows:

- In the context of archery coaching, estimate archery learners' conditions including emotion, stress and tiredness by using video and heartrate data.
- Reveal that IAC is useful as a variable that evaluates ERA in condition estimation.
- Propose a method to reduce the precision loss due to the bias between objective input data and subjective label data in condition estimation, where subjective self-reports are used as the ground truth.

2 RELATED WORK

2.1 Condition Estimation

Emotion estimation is a part of condition estimation. There are many related works on the topic of emotion estimation based on a variety of data [4, 5, 9]. However, biometric data that can be collected during archery practicing are limited. Considering the applicability to archery sport, we estimated condition using heart rate data from wearable devices and videos during the archery practicing. There are several models of emotion, including categorical and dimensional models [19]. In previous study, optimal zone on the dimension of positivity-negativity was identified. [20] Thus, in accordance with the previous study, in this study we used the PAD model, a 3-dimensional emotion model, which denotes emotions in pleasure, arousal and dominance dimensions. The pleasure(also called valence) dimension describes the positivity or negativity of emotion. The arousal dimension represent the level of excitement of emotion. The dominance dimension denotes the degree of feeling unrestricted to act in a variety of ways [14]. Furthermore, from the perspective of condition estimation for archery coaching, we consider factors that affect archery scores and that are important in archery coaching, to be a part of condition. From this perspective, tiredness and stress are also estimated as a part of condition.

2.2 Emotion Recognition Ability

The previous tests of emotion recognition ability(ERA) mainly focus on how precisely subjects can recognize emotions through facial or vocal expressions [10, 13]. We attempt to measure subjects' ERA

objectively from the viewpoint of subjects' perception ability of their own emotional state, by measuring interoception.

The interoception is a sense of the internal state of the body proposed by Charles Sherrington, a British physiologist [22]. IAC is the objective accuracy of the interoception. Barrett et al. revealed that people with higher IAC, emphasize their reports on the arousal of emotion experience more [1]. Terasawa et al. revealed that the IAC measured by heartbeat counting task(HCT) is highly related to ERA of facial expression [23]. HCT is a widely used method measuring IAC, with high test-retest reliability [6, 15]. The high test-retest reliability means that without training of heartbeat perception, the IAC evaluated by HCT of subjects does not varies dramatically. This study evaluated the IAC by HCT, and made use of the error score of HCT as the evaluation of subject's ERA. The error score of once HCT experiment is able to be used in a long-term.

3 COLLECTING DATA THROUGH EXPERIMENT

The data collecting experiment consisted of two parts and the HCT experiment was performed before the archery practicing simulation experiment for each subject.

3.1 Archery Practicing Simulation Experiment

In accordance with our research purpose that estimate condition for archery learners, 10 archery beginners were recruited in the archery practicing experiment.

Due to the limitation of the experiment environment, subjects were instructed to imitate the movement of practicing archery and to repeat the archery movement 10 times, 15s for each trial. An Apple Watch were worn on subjects' left arm, and a video camera was set on the right sides. The subjects' heart rate data was measured by the Apple Watch's photoplethysmography sensor and recorded by the Aware Framework [7, 16]. We used S.A.M(Self-assessment manikin) and VAS(Visual Analogue Scale) as our scale. S.A.M is an imagery-based scale for measuring emotion proposed by Lang, and is widely used in emotion estimation researches [2]. Tiredness and stress are measured by VAS (Figure 2), which is useful as stress scale [11, 12] and was recommended by JSFS (Japan Society of Fatigue Science) [17, 18].

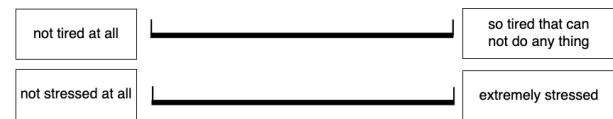


Figure 2: VAS Scale of Tiredness and Stress. The scales used in the experiment are in Japanese, the local native language.

3.2 Heartbeat Counting Task Experiment

The HCT is an experimental method to evaluate the subject's IAC by calculating the difference between accurate heartbeat and the heartbeat counted by subject. Based on Schandry's HCT experiment [21], we modified the test to make the test available on Apple Watch. During the experiment, subjects were seated in a quiet environment. We used apple watch to obtain a approximated single-lead electrocardiogram(ECG) for 30 seconds. Subjects were instructed

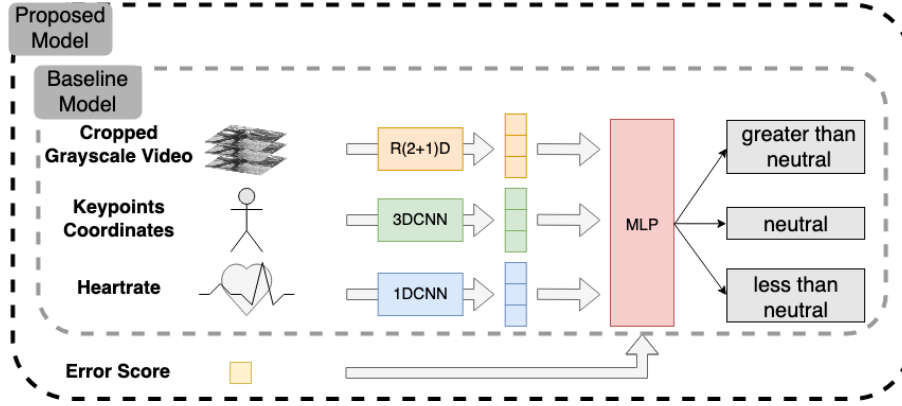


Figure 1: Baseline Model and Proposed Model. Compared with the baseline model, proposed model input Error Score additionally, which evaluates subject's individually different Emotion Recognition Ability

to count their heart rate during without touching their bodies. This 30-second experiment was repeated 5 times for each subject. Error score is calculated from monitored average heartbeat (N_{actual}) and subjects' report heartbeat (N_{report}). The average of 5-time error score is the explanatory variable that evaluate subject's individual ERA, in condition estimation model.

$$Error_score = \frac{|N_{actual} - N_{report}|}{N_{actual}} \quad (1)$$

4 CONDITION ESTIMATION MODEL

In this study, we designed a proposed model that included an explanatory variables evaluating ERA, and a baseline model that did not. The video and heartbeat data were preprocessed to output human joint coordinates(keypoints coordinates), grayscale video in the bounding box, and standardized heartrate data.

4.1 Data Preprocessing

4.1.1 Data Preprocessing of Videos. OpenPose is a real-time multi-person system capable of detecting human poses [3]. OpenPose can figure out the area including human(bounding box) and output coordinates of joints called as keypoints coordinates. We detected 17 keypoints using OpenPose. Videos were down-sampled from 300 frames to 16 frames at regular intervals. Believing that changes in keypoints coordinates can objectively represent body movements, keypoints coordinates were standardized and input into the model. We standardize 17 keypoints coordinates, while making standardized keypoints always have the distance between the neck and pelvis as 1, and the spine always overlaps the horizontal x-axis. To extract features from videos, grayscale videos cropped to the coordinates of bounding box were input to R(2+1)D model. R(2+1)D model is a ResNet (Residual Network) developed for action recognition from video, which combining 2DCNN for spatial analyzing and 1DCNN for temporal analyzing [24]. Because of memory space limitation, the video data was stored in grayscale. Grayscale videos were standardized and expanded to 3 channels, and processed to produce a 16-frame video with a size of 112×112 .

4.1.2 Data Preprocessing of Heartrate Data. Heartrate data detected from Apple Watch are not in a form of regular time intervals.

Therefore, we utilized cubic spline interpolation on the heartrate data. The interpolated data at 1-second intervals were standardized using Max-value to unify the scale of input data in machine learning models. 30 seconds of heartrate data were used as input data for the model.

4.2 Baseline Model and Proposed Model

The baseline model is a simple multimodal machine learning model as shown in the figure 1. The 3DCNN model was applied to the keypoints coordinates, and the 1DCNN model to the heartbeat data. We noticed that the middle option "neutral" of S.A.M is naturally distinguished from others. Therefore, the answers were divided into three classes: "neutral", "greater than neutral" and "less than neutral". We converted the numerical raw data of stress and tiredness into three categories similarly. The MLP was built from only one hidden layer. The learning rate was set to 0.005. Cross-entropy error and SGD was used as the loss function and optimization algorithms.

There was no difference between the structures of the proposed model and the baseline model, but the proposed method added the error score of the HCT as an explanatory variable, as shown in Figure 1. Error score was not passed through any layer, but was combined with other vectors and subsequently passed to a MLP model. As with the baseline model, the proposed model was 3-class model with the learning rate adjusted to 0.005, loss function set as cross-entropy error SGD used as optimization algorithms.

5 RESULT

The training data and test data are random separated from the original dataset at a ratio 8:2. Both models were trained on the training data, and were evaluated on test data. Since both models are multi-class classification models, in addition to accuracy, macro F1 scores are used as evaluation index.

The results of baseline model and proposed model are shown in Table 1. We evaluated both models in 5 dimensions: pleasure, arousal, dominance, tiredness and stress. Except the pleasure dimension, proposed model using IAc evaluated by HCT, showed better performance on the estimation of other 4 dimensions, which revealed the usefulness of proposed method. The improvement

of proposed method in arousal dimension is little but can not be ignored. In the dimensions of dominance, tiredness and stress, proposed method showed approximately 10% increasement of both accuracy and macro F1 score.

Table 1: Results of 2 Methods

	Baseline		Proposed	
	Accuracy	Macro F1	Accuracy	Macro F1
Pleasure	0.85	0.813	0.80	0.736
Arousal	0.60	0.561	0.60	0.599
Dominance	0.50	0.491	0.60	0.611
Tiredness	0.50	0.460	0.60	0.581
Stress	0.70	0.513	0.80	0.656

6 DISCUSSION

We want to have a discussion about the reason why the pleasure dimension differs from the four dimensions of arousal, dominance, tiredness and stress, did not show better performance in proposed method.

Wiens et al. revealed that their experimental results were consistent with the hypothesis that the subject's self-report of the valence dimension, which represents the subject's pleasantness or unpleasantness degree, is not related to his or her IAc [25]. Furthermore, according to the previous research by Barrett et al., the arousal focus of experienced emotion, which is the amount of information about arousal dimension contained in verbal reports, revealed to be related to IAc, but valence focus, the similar term in valence dimension, not [1].

Referring to previous studies, the IAc of subject, measured by error score in this study, is not related to the subjective valence self-report. The lower estimation accuracy of the proposed model for emotional valence reveals that the IAc may not be suitable to evaluate the ability to recognize valence.

The proposed method was inferior to the baseline method for estimating the valence dimension. It is necessary to identify the variables that can evaluate the ability of valence recognition in the future.

This study attempted to reduce the bias between subjective self-reports and objectively detected data in condition estimation. We conducted the experiment of 10 subjects, which is a small sample. Experiment in larger scale is expected in the future. While the usefulness of the method was demonstrated in the estimation of conditions during the simulation of archery, future work is needed to clarify the usefulness of the method in general-purpose condition estimation.

In addition, several methods for evaluating the IAc have been proposed, but all of the methods were conducted in a laboratory environment, but not a general environment. Future research on methods to evaluate the IAc suitable for condition estimation is also an issue to be addressed.

Some concerns are about the practicality of the proposed method, because the procedure of HCT is so complicated to repeat every time. However, it is a unnecessary concern, because as have mentioned above, the IAc evaluated by HCT is with a good test-retest reliability, so can be used in a long term.

7 CONCLUSION

In this study, we conducted an experiment to estimate the condition of archery learners in 5 dimensions: 3 dimensions of the PAD model, and 2 dimensions of tiredness and stress. Compared to a baseline model with only video data and heartrate data as input, the proposed method, in which the error score of HCT was added as a variable to evaluate the ERA, improved the accuracy and Macro F1 score in 4 dimensions: arousal, dominance, tiredness, and stress. The reasons why the proposed method did not more accurately estimate the valence dimension and the future issues were discussed.

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